

# BAAZ JKBOSE



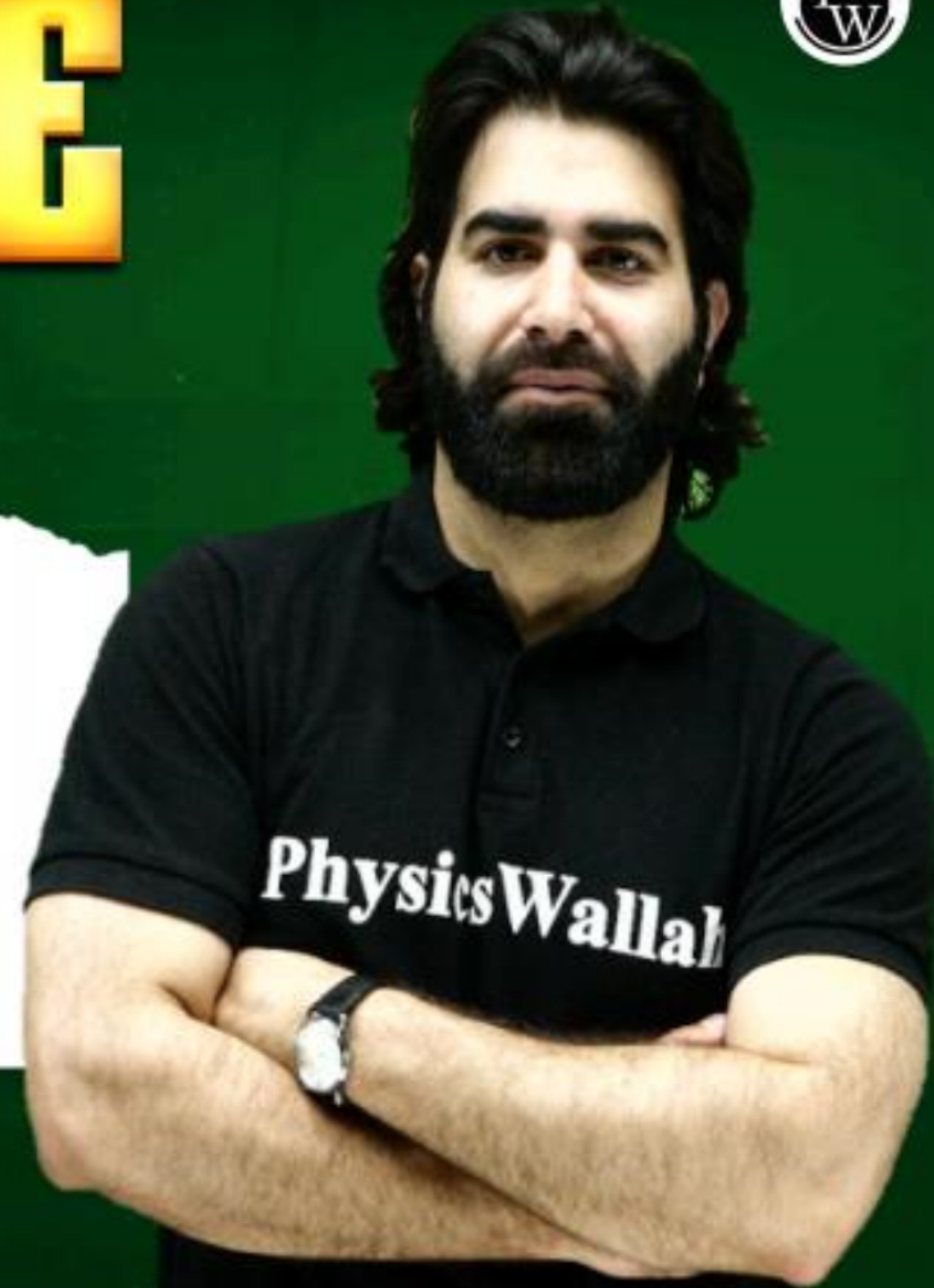
2026 CLASS-11<sup>TH</sup>

Locomotion and  
Movement

LECTURE- 01

Zoology

Dr Adil Altaf sir



# Topics To Be Covered

1

Introduction, types of movement, types of muscles

# Revision

↳ Composition of Urine

→ H/w

→ Renal Calculi

→ Hard Masses / Crystalline Masses

(Kidney Stone)

→ Mainly Made of Calcium oxalate, Uric Acid etc

Salts



→ Stone

→ Glomerulo Nephritis

→ Inflammation of Glomerulus of kidney

- klas Bright's Disease

- RBC's in Urine

## Movement & Locomotion

Movement - change in the Body Posture.

or  
change in the position of Body organs without changing the place.

Locomotion → When the Movement causes change in the place/position the organism was present



"Every locomotion is a movement but Every Movement is not a locomotion".

## Types of Movement (Animals)

e.g. → Jellyfish, Octopus etc

① Tenticular Movements - Due to Tentacles.



② Amoeboid Movement - Due to Pseudopodia

e.g. Amoeba.

- NBC, (Macrophages, Neutrophils etc).



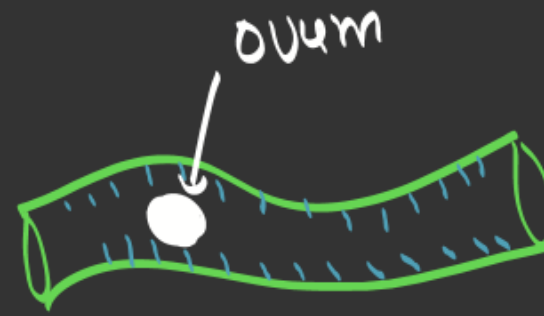
③ Ciliary Movement - Due to Cilia.



e.g. - Paramecium.

- Fallopian tubes.

- Respiratory tract.



④ Flagellated Movement - Due to Flagella.



e.g. → Trypanosoma.

→ Sperm cell.

⑤ Muscular Movement

- Due to Muscles

e.g. Human Skeletal Muscles

## - Muscle →

↳ Muscle or Muscular tissue is one of the Basic Tissues in Animals.

↳ Mesodermal is origin.

↳ Muscles have the property -

- ✓ Contraction [Contractibility]
- ✓ Elasticity [Can Regain their Shape]
- ✓ Extensibility [Can be Extended]
- ✓ Excitability [Show Response to Signal]

↳ There are 3 types of Muscles in Humans

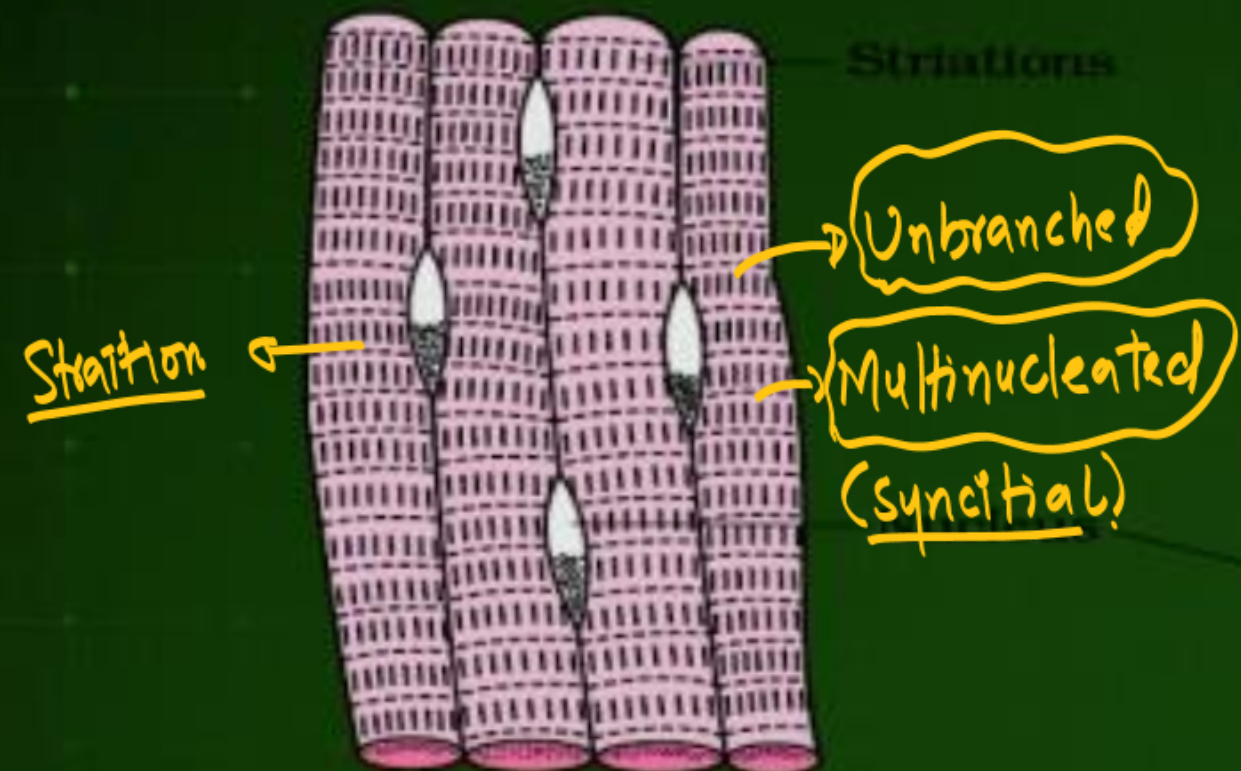
① Skeletal "

② Cardiac Muscle

③ Smooth Muscle.

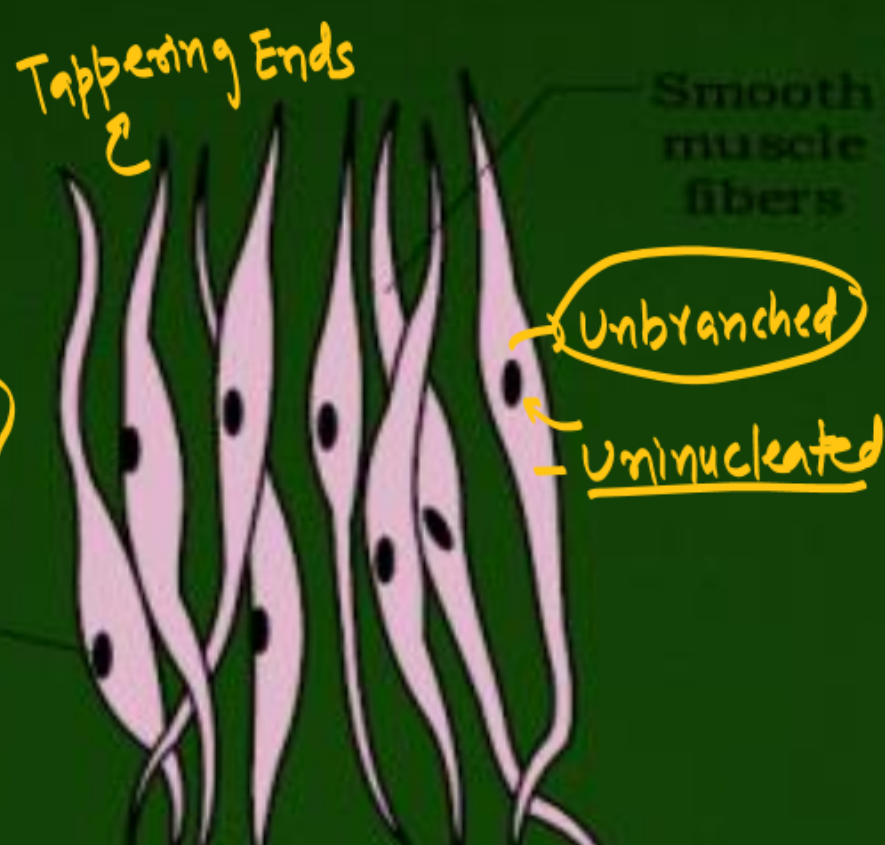
Others

- Nervous tissue
- Epithelial tissue
- Connective tissue



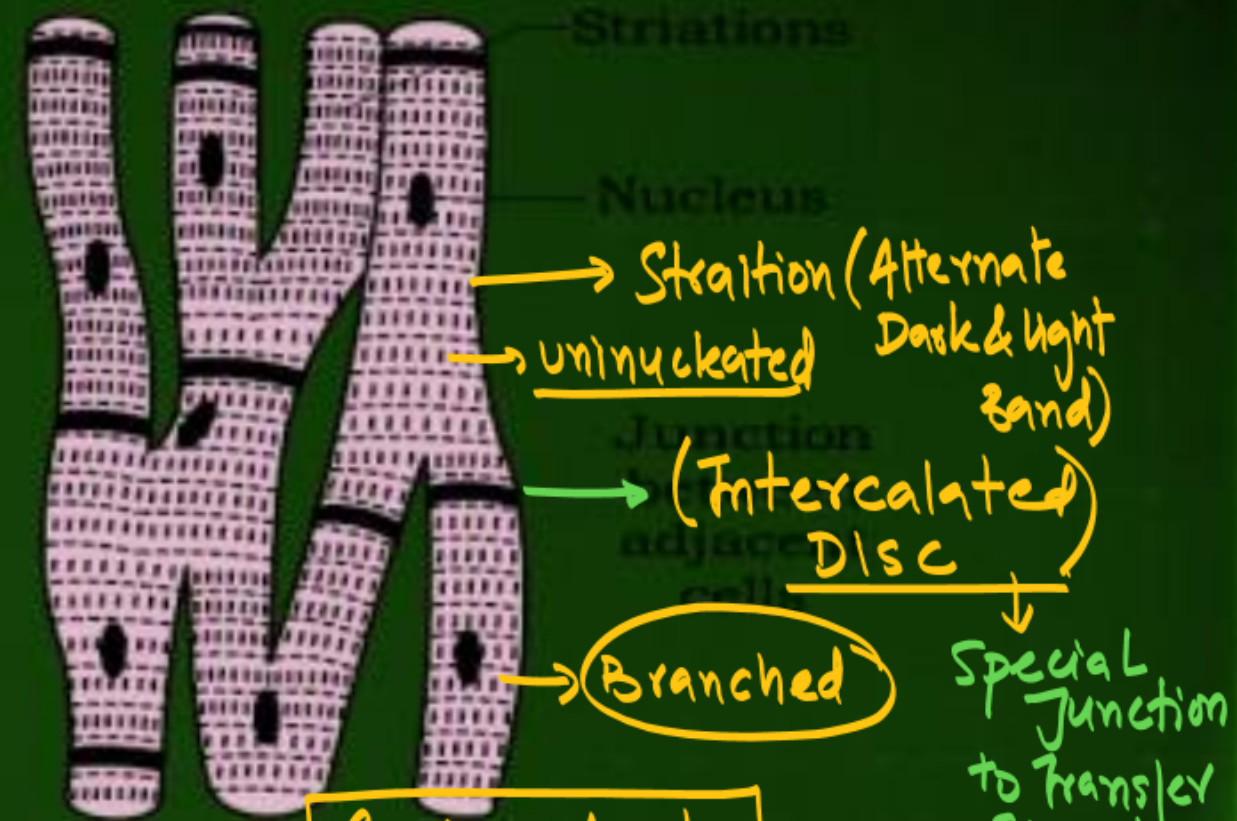
**Skeletal Muscle**

- ↳ Associated with our skeleton/Bones
- ↳ **Voluntary** (under own control)
- ↳ **Striated** - (have striation)



**Smooth Muscle**

- ↳ Called so, because it appears smooth under microscope
- ↳ It is **Non-Striated**
- ↳ Associated with visceral organs (internal)
  - e.g. - Stomach, Intestine, U. Bladder, Blood vessels etc.
- ↳ **Involuntary**
- ↳ **Non-Striated**

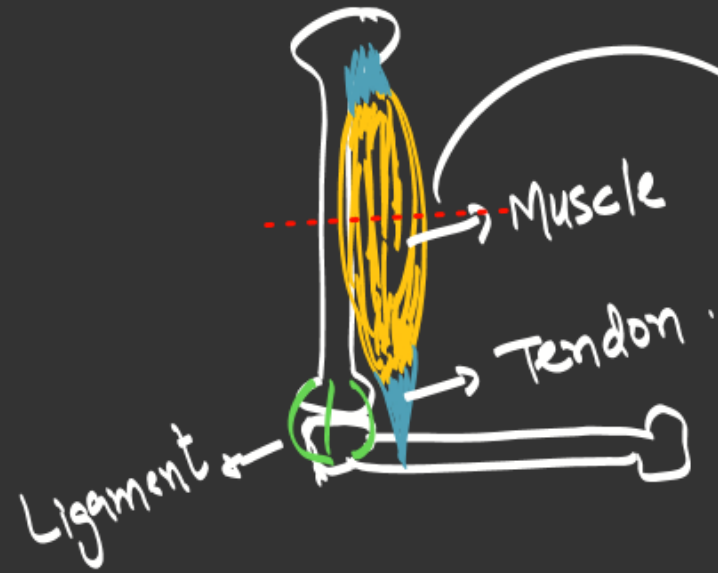


**Cardiac Muscle**

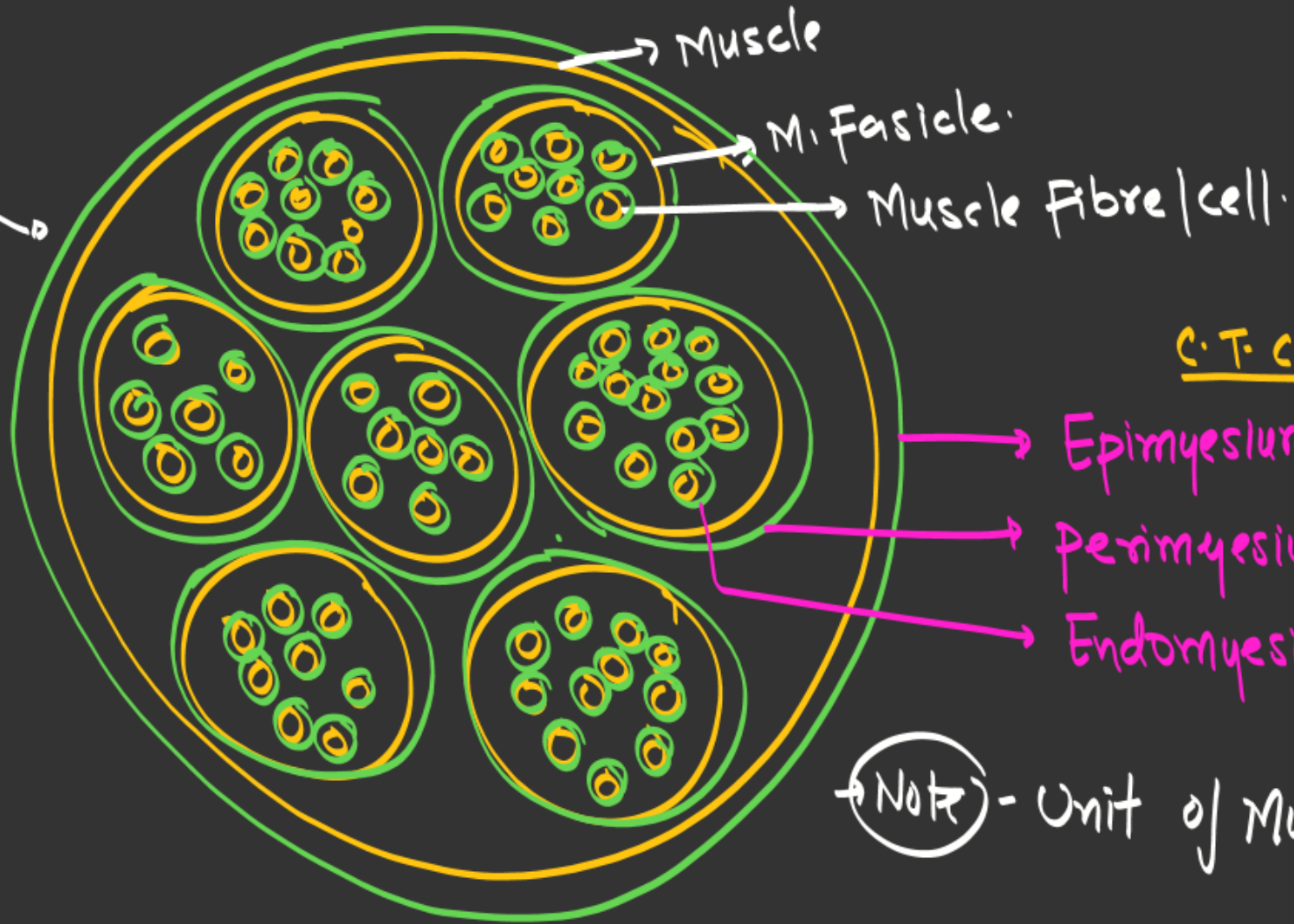
- ↳ Exclusively present in heart
- ↳ **Involuntary** [Not under own control]
- ↳ **Striated**

## Skeletal Muscle

- ↳ A Skeletal Muscle is Made of Muscle Fascicles/Bundles.
- ↳ A Muscle Fascicle is Made of Muscle Fibre/Cell.
- ↳ A Muscle Fibre is Actually a Muscle cell. [As it is elongated, hence called Fibre].



Section



(Tendon) - Connects Muscle to Bone.

- Ligament - Connects Bone to Bone (Joint).

C.T. covering (Fascia)

Epimysium [on Muscle]

Perimysium [on Fascicle]

Endomysium [on Fibre/cell]

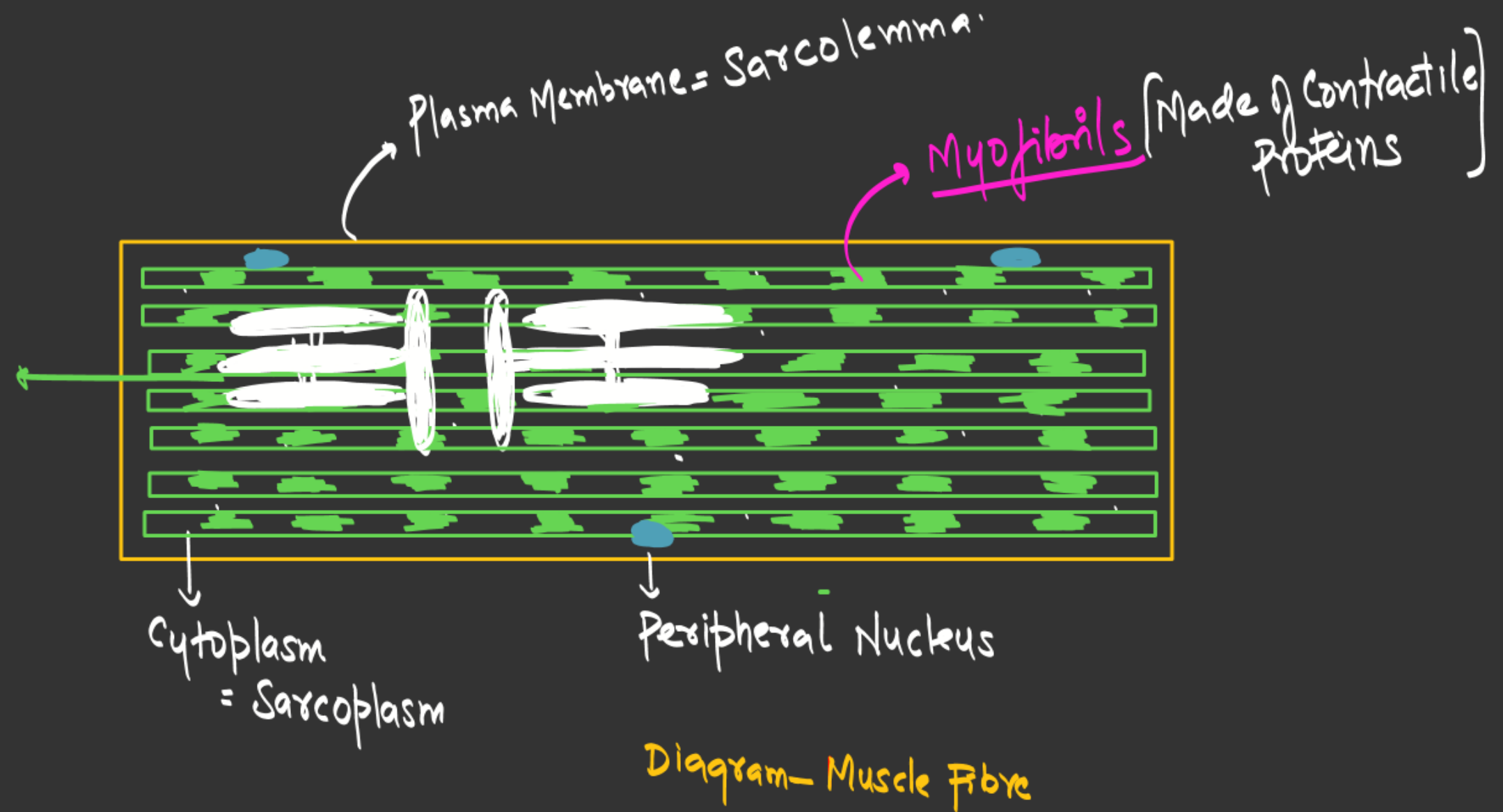
Note - Unit of Muscle is Muscle Fibre.

# Muscle Fibre / cell.

↳ Also klas Myocyte.  
↓  
Muscle cell.

Muscle  
↓  
Muscle fascicle  
↓  
Muscle Fibre  
↓  
Myofibril.

Endoplasmic Reticulum  
[Sarcoplasmic Reticulum]  
↳ Stores  $Ca^{++}$



Thank  
You